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CO1-AT3

1.Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Cloud computing is the delivery of computing services (servers, storage, databases, networking, software) over the internet on a pay-as-you-go basis.

Any 4 Characteristics

1. **On-Demand Self-Service**
   Users can access resources automatically without provider interaction.
2. **Broad Network Access**
   Services are available over the internet on multiple devices.
3. **Resource Pooling**
   Multiple users share the same infrastructure (multi-tenant model).
4. **Rapid Elasticity**
   Resources can scale up or down quickly based on demand.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Hardware Evolution

- Mainframes → Minicomputers → PCs → Client-Server → Distributed Systems → Data Centers
  Enabled powerful, shared computing infrastructure

Software Evolution

- Standalone apps → Client-server apps → Web apps → SOA → Cloud-native apps
  Enabled internet-based service delivery

Conclusion

These evolutions led to:

- Virtualization
- High-speed internet
- Scalable cloud platforms

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Definition

Server virtualization divides one physical server into multiple virtual machines using a hypervisor.

Role in Cloud

- Improves resource utilization
- Enables scalability
- Supports multi-tenancy

Difference

Virtualization	**Cloud Computing**
Technology	Service model
Creates VMs	Delivers services
Works at hardware level	Works at service level

4. Discuss the role of data centers in delivering cloud services..

Definition

A data center is a facility with servers, storage, and networking used to deliver cloud services.

Role

- Hosts cloud infrastructure
- Ensures high availability
- Provides security and backup
- Supports global service delivery

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Model	**Description**	**Example**
IaaS	Provides virtual hardware	AWS EC2
PaaS	Platform for app development	Google App Engine
SaaS	Ready-to-use software	Gmail
XaaS	Everything as a service	AWS Cloud